Chapter 6

Nothing but Music: The Essentials of Superstring Theory

Music has long since provided the metaphors of choice for those puzzling over questions of cosmic concern. From the ancient Pythagorean “music of the spheres” to the “harmonies of nature” that have guided inquiry through the ages, we have collectively sought the song of nature in the gentle wanderings of celestial bodies and the riotous fulminations of subatomic particles. With the discovery of superstring theory, musical metaphors take on a startling reality, for the theory suggests that the microscopic landscape is suffused with tiny strings whose vibrational patterns orchestrate the evolution of the cosmos. The winds of change, according to superstring theory, gust through an aeolian universe.

By contrast, the standard model views the elementary constituents of the universe as pointlike ingredients with no internal structure. As powerful as this approach is (as we have mentioned, essentially every prediction about the microworld made by the standard model has been verified down to about a billionth of a billionth of a meter, the present-day technological limit), the standard model cannot be a complete or final theory because it does not include gravity. Moreover, attempts to incorporate gravity into its quantum-mechanical framework have failed due to the violent fluctuations in the spatial fabric that appear at ultramicroscopic distances—that is, distances shorter than the Planck length. The unresolved conflict has impelled a search for an even deeper understanding of nature. In 1984, the physicists Michael Green, then of Queen Mary College, and John Schwarz of the California Institute of Technology provided the first piece of convincing evidence that superstring theory (or string theory, for short) might well provide this understanding.

String theory offers a novel and profound modification to our theoretical description of the ultramicroscopic properties of the universe—a modification that, physicists slowly realized, alters Einstein’s general relativity in just the right way to make it fully compatible with the laws of quantum mechanics. According to string theory, the elementary ingredients of the universe are not point